Oversampling rare failures: what SMOTE does to surrogate-based failure probabilities

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Abstract. Classifiers are increasingly used as surrogates of limit-state functions in structural reliability, and because failures are rare they are often trained on data rebalanced with the synthetic minority oversampling technique (SMOTE). Its effect on the estimated failure probability P_f has not, to our knowledge, been measured. We compare, on three standard benchmarks (a four-branch series system, a non-linear oscillator and a ten-variable linear limit state), a classifier trained on plain Monte Carlo samples, class weighting, SMOTE, and SMOTE followed by a Bayes prior-shift correction, with gradient boosting and a neural network, 50 repetitions each and 2000 training points. The direction of the error depends on how many failures the training sample contains. With about nine failures (four-branch, $P_f = 4.4 \times 10^{-3}$) SMOTE overestimates P_f by a median factor of 11–13; the correction reduces this to 0.6–2.4. With about 57 failures (oscillator) all strategies are within a factor 0.7–1.2. With about three failures (linear, $P_f = 1.35 \times 10^{-3}$) the plain classifier collapses to medians of 0.00–0.03 of the true value, SMOTE recovers 0.6–0.7 but with a 10–90% range of 0.12–1.96, and the correction collapses again. No strategy is safe across regimes: a thresholded classifier, rebalanced or not, should not be used as a failure-probability estimator without a check against the true limit state.

1. Introduction

Surrogate models make reliability analysis affordable when each evaluation of the limit-state function g is expensive [1]. Classification surrogates predict the sign of g, and the failure probability is estimated by applying the classifier to a large Monte Carlo sample. Because failures are rare, the training data are imbalanced, and rebalancing with SMOTE [2] has been proposed to help the surrogate learn the failure region, for example in ship-structure design [3]. In risk prediction for medicine, rebalancing is known to miscalibrate predicted probabilities [4]. Whether the same happens to P_f in reliability analysis, and in which direction, is the question of this note. It was prompted by an automated forecast that SMOTE would enter computational mechanics.

2. Methods

Benchmarks (standard normal inputs).

- Four-branch series system [5], 2 variables, $k = 6$: $P_f = 4.44 \times 10^{-3}$ (from 10^6 samples).
- Non-linear undamped oscillator, 6 variables, with the usual parameter distributions: $P_f = 2.85 \times 10^{-2}$.
- Linear limit state $g = 3\sqrt{10} - \sum_{i=1}^{10} x_i$: $P_f = \Phi(-3) = 1.35 \times 10^{-3}$ exactly.

Strategies. A classifier is trained on 2000 Monte Carlo points labelled by the sign of g: (i) plain; (ii) class weights inversely proportional to class frequency (gradient boosting only); (iii) SMOTE to a

balanced sample ($k = 5$); (iv) SMOTE followed by the prior-shift correction $p' = \frac{p\,\hat{\pi}/\pi_s}{p\,\hat{\pi}/\pi_s + (1-p)(1-\hat{\pi})/(1-\pi_s)}$, where $\pi_s = 0.5$ is the balanced prevalence and $\hat{\pi}$ the failure share in the training sample. P_f is estimated as the share of 2×10^5 new points classified as failures ($p > 0.5$).

Classifiers. Histogram gradient boosting (200 iterations) and a two-layer perceptron (32–32).

Protocol. 50 repetitions per configuration; repetitions with fewer than two failures in the training sample are skipped and counted (13–14 of 50 for the linear case). Everything was fixed before the single run.

3. Results

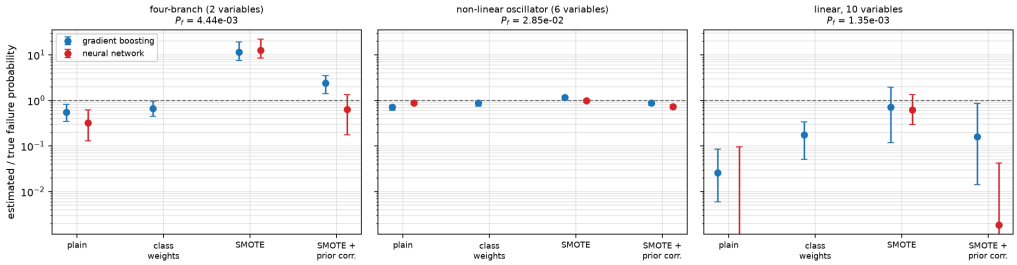


Figure 1. Estimated over true failure probability: median and 10–90% range over repetitions (log scale).

Table 1. Median ratio of estimated to true P_f.

benchmark	classifier	plain	class weights	SMOTE	SMOTE + correction
four-branch	boosting	0.56	0.67	11.4	2.39
	neural net	0.32	–	12.8	0.63
oscillator	boosting	0.72	0.88	1.17	0.88
	neural net	0.88	–	0.99	0.74
linear, 10 variables	boosting	0.03	0.17	0.71	0.16
	neural net	0.00	–	0.63	0.00

Three regimes appear (Figure 1, Table 1).

- **Few but informative failures** (four-branch, about 9 in the training sample): the plain classifier underestimates (0.3–0.6), SMOTE overestimates by an order of magnitude, the correction brings the estimate back within a factor 2.5.
- **Many failures** (oscillator, about 57): every strategy is within a factor 0.7–1.2; SMOTE is the closest.
- **Very few failures** (linear, about 3): the plain classifier predicts almost no failures; SMOTE is the only strategy near the true value, but with a spread of more than an order of magnitude; the correction undoes it.

SMOTE always raises recall (for example from 0.35 to 0.73 on the four-branch system), which is why it is attractive; but recall is not the quantity a reliability analysis needs.

4. Discussion

SMOTE moves the decision boundary toward the safe region. Whether this corrects or creates bias depends on how far the plain classifier was from the true boundary, which is unknown in practice. The prior-shift correction is exact for calibrated probabilities but cannot fix a boundary that was never learnt. The practical rule is therefore negative but useful: report a classifier-based P_f only together with a check against the true limit state — for example evaluating g on the points the surrogate classifies as failures, or using the surrogate only to build an importance sampling density — and prefer adaptive designs that place training points near the boundary.

Limitations. Three benchmarks, one training size, two classifiers, a 0.5 threshold; adaptive (active learning) surrogates, which are the state of the art, are not compared.

Reproducibility. `experiment.py` regenerates every number and the figure.

Origin of the question. The question came from an automated forecast (Singularity v3, registered on 5 November 2026 with a SHA-256 fingerprint, view without generative AI) that SMOTE would enter the subfield *Computational Mechanics*. The forecast concerned the arrival of the method, not the result of this note.

References

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